

Galaxy alignments are uncertain across spatial scale and time

Restricted response — NLA example

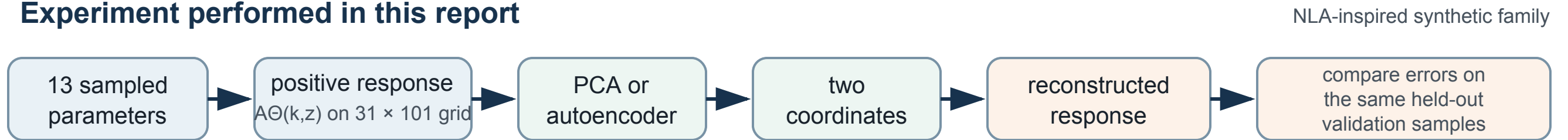
- fewer adjustable parameters
 - easier to constrain and explore
 - may miss relevant behavior
 - can shift inferred cosmology if inadequate
- can be adequate for some data and scales

More flexible response — TATT example

- additional tidal-response terms
 - wider family of possible responses
 - more weakly constrained directions
 - extra computation and parameter degeneracies
- more flexibility is not automatically better

Can a flexible response family be described with fewer numbers at useful accuracy?

Experiment performed in this report



Still required before cosmological use

errors in lensing predictions • probabilities for new coordinates • tests of unbiased inference